

Python for QA - Installation & Setup

Readable programming language with a strong ecosystem for automation, APIs, data and utilities.

Goal of this manual

This manual guides you through the complete installation and configuration process, from preparing the environment to final validation. Follow the steps in order and use each verification checkpoint to confirm that the setup was completed correctly.

Before you start

Confirm the supported operating system, runtime/dependencies and version requirements in the current official documentation before installing. The technical checklist below mirrors the validated tool-specific procedure.

Where to run commands

On Windows use PowerShell or Command Prompt; on macOS/Linux use Terminal. After changing PATH or environment variables, close and reopen the terminal before validating again.

Official references

https: [//www.python.org/downloads/](https://www.python.org/downloads/)

Step-by-step installation and setup

1. Follow this tool-specific step: Acesse [python.org/downloads](https://www.python.org/downloads/) e siga a opção oficial recomendada para seu sistema.
2. Follow this tool-specific step: No Windows, durante a instalação/gerenciador, habilite a disponibilização do Python no PATH quando essa opção for apresentada.
3. Follow this tool-specific step: Abra um NOVO PowerShell e execute `python --version`.
4. Follow this tool-specific step: No Windows, execute também `py --version`.
5. Follow this tool-specific step: Valide pip com `python -m pip --version`.
6. Follow this tool-specific step: Crie uma pasta: `mkdir python-lab` e entre nela.
7. Follow this tool-specific step: Crie ambiente virtual: `python -m venv .venv`.
8. Follow this tool-specific step: PowerShell: `.venv\Scripts\Activate.ps1`; CMD: `.venv\Scripts\activate.bat`; macOS/Linux: `source .venv/bin/activate`.
9. Follow this tool-specific step: Confirme que o terminal indica o ambiente `.venv` e execute `python -m pip --version` novamente.
10. Follow this tool-specific step: Crie `hello.py` com `print("QA Lab Brasil - Python funcionando")`.
11. Follow this tool-specific step: Execute `python hello.py`.

Commands and validation

```
python --version
```

```
py --version
```

```
python -m pip --version
python -m venv .venv
.venv\Scripts\Activate.ps1
python hello.py
```

First working use

Complete the first working use described in the steps above and confirm the expected tool-specific result before adding integrations or advanced configuration.

Common problems and fixes

python não reconhecido, mas py funciona: o runtime existe; use py temporariamente e revise os aliases/PATH conforme a documentação atual do Python para Windows.

pip não reconhecido: prefira python -m pip em vez de depender do comando pip global.

Ativação .venv bloqueada no PowerShell: use CMD/activate.bat ou siga a política de execução aprovada pela organização.

Múltiplos Python no PATH: execute where python no Windows ou which python no macOS/Linux e remova/ajuste entradas antigas com cuidado.

Completion criteria

Treat setup as complete when the version/execution checks work and the first use in this manual finishes without errors. Only then move to integrations or advanced configuration.